

The Mechanics of Effective Planning in Transformer Models

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Abstract - Planning with Reasoning Agents in domains like Software Engineering involves searching through different files, evaluating possible changes, and sometimes communicating with other agents. Topologically, the resulting context used for planning resembles a graph navigation problem. Where given an exploration trace between two points, the model needs to discard dead-end subpaths, compare competing viable alternatives, and preserve the graph connectivity to find the shortest path. The synthetic task allows measuring the planning capacity of transformers decoupled from semantic memorization. We compare accurate versus inaccurate plans with respect to three axes: the entropy of attention maps during inference, the topology of the graph-exploration trace, and the path-generation algorithm. Results show that, in good plans, the last layer allocates ≈ 5.98 more attention to the correct path. Additionally, we demonstrate that effective planning is influenced by the position, depth, and frequency of bifurcations, while random walks are more challenging than deterministic algorithms. Applications of the research include automatic plan verification, enhanced reasoning trace generation during training or inference, and theoretical foundations to develop a superior linear attention model.

Good plans exhibit lower entropy

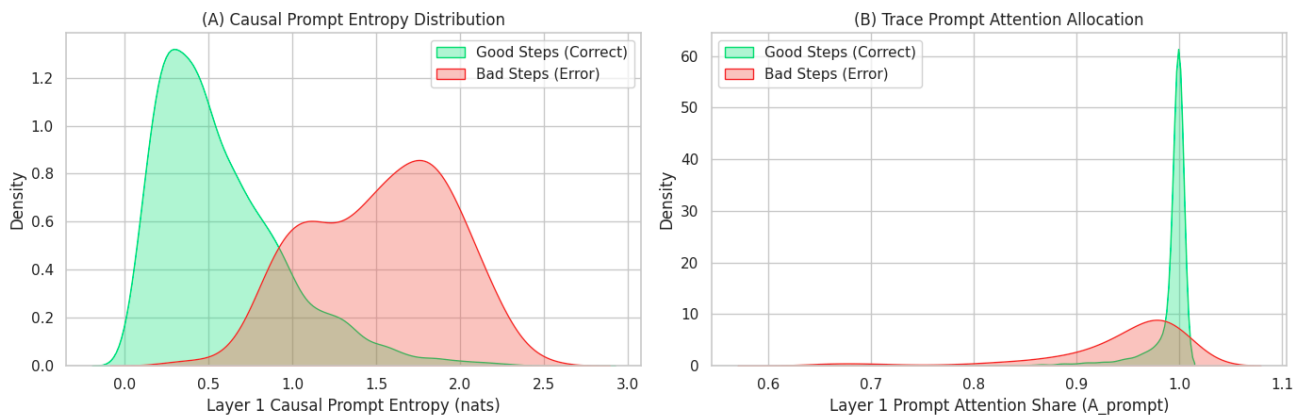


Figure 1 - Causal Self-Attention Mechanics in Good vs Bad Plans

Methodology - To compare good plans versus bad plans we utilize a checkpoint trained after 100 epochs that achieved 75.40% exact-path accuracy, and 97.34% token accuracy in the validation set [\[notebook 3.1-cell 5\]](#). The model achieved 91.40% exact-path accuracy at epoch 200, and 99.40% accuracy at epoch 100. The results from comparing good versus bad plans were validated by comparing inference at epoch 100, and epoch 1000. The model used was a 2 layer decoder-only transformer architecture with 2 attention heads for each layer, 32 embedding dimensions, and 64 hidden dimensions. In total, the model had 19,818 parameters [\[3.1-cell 3\]](#). The dataset contained 3000 training samples, and 1,000 total for validation, and testing - both used for the analysis. The vocabulary size was 40 tokens to represent nodes, and 2 extra tokens for padding, and stopping. The exploration trace (input sequence) had a length between 30 & 50 tokens. While the shortest-path (output) was between 10 & 20 tokens. The exploration trace followed a depth-first search algorithm [\[0.1-cell 2\]](#).

Results - The last attention layer shows that good plans exhibit lower entropy, and better allocation of attention. 12,005 decision-steps were evaluated with errors on 306 of them. Errors occurred 83.66% at bifurcations, defined as a node that appeared at least twice in the exploration trace. Accuracy in bifurcations was 92.25% (256 errors in 3304 steps), compared to 99.43% in single steps (50 / 8701) [\[4.3-cell 5\]](#). The main causal predictor of good predictions was the percentage of attention allocated to the last occurrence of the present token in the exploration trace - which in a bifurcation, indicates what path to follow. Good plans show that on average 13.39 more attention is allocated to the last occurrence of the token - the correct one - based on the DFS

path-generation algorithm compared to only 2.24 more in bad plans [4.3-cell 7]. Entropy in the last attention layer translates to higher confidence measured by the delta between the top tokens when making a prediction in the classification layer of the model. For a similar encoder-transformer model it showed a 4.02 margin in good plans versus a 3.05 margin in bad plans [2.0-cell 5]. As training progressed in the decoder-only model the margin of confidence increased from 6.89 at epoch 100 to 10.07 at epoch 1,000 (considering bifurcations, and single steps) [4.2-cell 4]. Figure 1.A shows the entropy distribution across all tokens in the final attention layer, and figure 1.B shows the comparative difference between the first, and last occurrence of the present token when making a prediction in a bifurcation [4.3-cell 9]. Future work may focus on understanding the circuit dynamics in preceding layers that enable the final attention layer to focus on the correct subpath (N. Elhage, 2021 - Anthropic). As well, as how attention evolves when increasing the depth of the model for more complex graphs, and path-generation algorithms.

Graph Topology influences effective planning

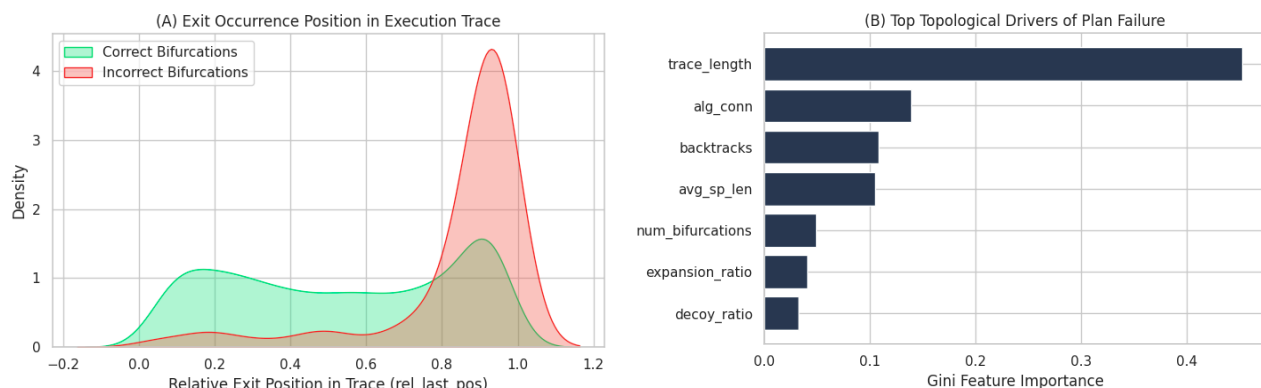


Figure 2 - Topological Contrast in Bifurcation Steps

Causal analysis - We consider the topology of the exploration trace as the primary source of planning errors. Analysis was done at the full trace, and at the particular bifurcation step where the error happened. For the full trace, length was the primary predictor of errors with 35.43 input tokens on average in good plans, and 42.91 in bad ones (p -value 2.55×10^{-33}). Second, the number of bifurcations: 8.63 in good plans, versus 11.49 in bad plans (p 1.69×10^{-28}). Decoy ratio, the percentage of nodes present in the trace that were not part of the shortest path, was the third most significant feature with 72.51% in good plans vs 63.99% (p 6.13×10^{-25}). Other significant features include average degree, and density (both at the graph level) and the number of backtracks: bifurcations with immediate U-turns in the exploration trace (5.63 vs 7.52) [4.4-cell 5]. Figure 2.B shows the Gini feature importance based on a gradient boosting model [4.4-cell 6].

Bifurcation Topology - At the individual step level, the position, frequency, and depth of the bifurcations were analyzed [4.3-cell 6]. In good predictions, the relative position of the first occurrence of the token representing the node in a bifurcation was 0.344 compared to 0.641 in bad plans (p 2.96×10^{-47}). For the last occurrence of the token in the exploration trace, the relative position was 0.537 in good plans versus 0.844 in bad ones (p 2.77×10^{-68}). Figure 2.B shows the distribution of the last occurrence, the correct subpath, in accurate, and inaccurate plans [4.3-cell 9]. Planning effectiveness also decreased based on the frequency of occurrences for the bifurcation token. When the node was visited twice accuracy was 93.45% (175 errors in 2,671 occurrences). For a frequency of three visits accuracy dropped to 88.68% (61 / 539), and for a frequency of four visits it dropped to 78.72% (20 / 94). The depth of the bifurcation, the trace steps between the first entry, and the last exit had an average of 7.16 steps in good plans, and 9.00 in bad plans with a p -value of 2.79×10^{-3} .

An ablation study could indicate if errors in long input sequences, and bifurcations near the end of the sequence are a result of a smaller number of training samples or a spurious correlation learned from the positional encodings of the average exit position for a bifurcation. However, the results are consistent with research that indicates that longer reasoning sequences tend to be less faithful (T. Lanham, 2023 - Anthropic). Future work may explore possible applications of improving language models based on the reasoning topology.

Random Walks are significantly more challenging

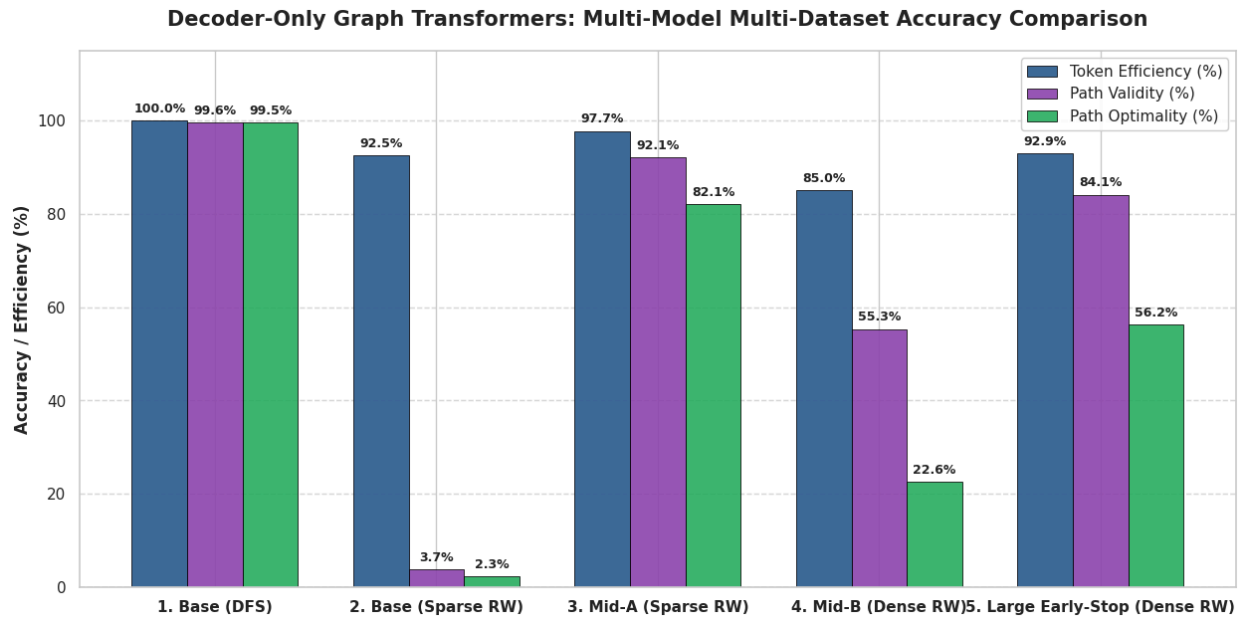


Figure 3 - Accuracy comparison across datasets, and models based on the path generation algorithm [3.5-cell 7].

Dataset Generation - finding the shortest path in a depth first search graph exploration trace may be facilitated by the fact that the correct path is always the last to appear, which can be easily modeled by a skip-trigram that transformers have been proven to achieve at a depth greater than 1. A more challenging dataset was designed with two key differences: increase the average node degree from 1.92 to 6.90, and use a random walk to generate the exploration trace [0.1-cell 4 & 0.3-cell 4]. The input, and output sequence length remained the same, as well as the vocabulary. The trace generation algorithm sampled multiple candidates that met length requirements in a random walk, and selected the most challenging based on the number of alternate paths (multiple viable solutions), cyclicality, clustering coefficient, and number of nodes in the trace not part of the trace (decoy rate) [0.3-cell 2]. Future work may explore how each characteristic independently contributes to planning difficulty.

Training results - Two flavors of random walk datasets were evaluated across three different model sizes. Results will be compared to models trained on the depth-first-search dataset, all after 1,000 epochs that achieved 99.40% optimal path accuracy with 19,818 parameters [3.1-cell 3]. The first dataset is not densely connected, but exploration is based on random walk. The main challenge, contrary to bifurcations, are back-and-forth steps significantly increasing the difficulty of previously single-step predictions. The small model with similar architecture (19,818 parameters) achieved only 1.60% exact match, 3.60% path validity, and token accuracy of 85.24% [3.2-cell 5]. A larger model with 139,306 parameters (4 layers, 64 embedding, and 128 hidden dimensions) achieved 80.80% exact match, 92.80% path validity, and 98.06% token accuracy [3.2A-cell 5]. For the dense dataset, a model with 72,362 parameters (2 layers, 64 & 128 dimensions) achieved 22.6% accuracy on the optimal path, 55.3% path validity, and 85.22% token accuracy [3.3-cell 5]. Results show that it performed better on the sparse dataset, despite being trained only on the dense datasets. A larger model with 540,714 parameters (4 layers, 128 & 256 dimensions) stopped early after 100 epochs showed path validity of 84.60% vs 49.20% (in the 72K model), exact match of 41% (vs 11.20%) and token accuracy of 93.25% (vs 84.78%) [3.3A-cell 5].

Related work shows that transformers achieve 97.2% accuracy on a shortest path task. However, it was trained for 1 million epochs, with a 60 million parameter model, and a slightly larger dataset (100k samples vs 30k). They also allocated specific tokens for edges, and provided the entire graph as input (C. Sanford, 2024 - Google).

Future Work

1.- Automatic plan verification: there is a growing need for alignment in Software Engineering. Enterprises increasingly depend on autonomous agents for code development, verification, and security. Errors in software agents may lead to production bugs, or data privacy breaches. Automatic plan verification enables selecting the best planning alternative, and reducing costs by only allocating development resources only to the best plans. The primary obstacle from translating insights in a synthetic task to software engineering is the length of the reasoning trace, and the far larger vocabulary size. Splitting the reasoning trace by tool usage would enable formulating the reasoning trace to a similar topology as in a graph. It would also enable measuring the certainty of a given tool usage. For example, the entropy when searching for a file, or when assigning a subagent with a task. This could improve tool usage: what tool to use, but also when to use a tool. If entropy is high, before assigning a task - perhaps, the planning agent requires more information. The resulting system could translate to other domains as deep research, multi-agent enterprise systems, and autonomous scientific discovery. The first steps would be comparing entropy in effective software planning trajectories versus inaccurate ones. Then, modeling the reasoning trace topology and identify the leading causes of error.

2.- Enhanced reasoning trace generation: results show that the topology of the input influences the ability to plan. Which leads to wonder if improving the topology of the reasoning trace may lead to improvements in accuracy for reasoning models. Applications include modifying the trace during training, and also during inference. Given a causal relationship between the topology of reasoning trace, and the accuracy of the model would lead to designing a better curriculum for training on reasoning for common topologies that exhibit bad accuracy. These could include the length of the reasoning sequence, the dispersion between related alternatives, the depth when evaluating a given alternative, and the path of the reasoning trace with an emphasis of back-and-forths between two competing options. The results would also lead to improving accuracy during training by using multi-agent systems to transform the topology of the original reasoning trace, or information gathering tasks in domains like software engineering to a more agreeable topology that leads to less complex reasoning. Initial efforts would also focus on evaluating the accuracy of the reasoning trace based on its topology. Probably, with a multi-agent system inspired by the architecture of Recursive Language Models that enables programmatically to evaluate a long-context in shorter sequences ([A. Zhang, 2026 - MIT](#)).

3. Theoretical foundations for a superior attention: the task of finding the shortest path has a few interesting properties for measuring, and developing superior attention mechanisms. For once, the task has a negative correlation between input length, and accuracy, given that the length of the shortest path remains fixed. Which implies there is noise, or disinformation in the context. In the more extreme case, if the input (exploration trace) is as long as the output (shortest path) the task is the easiest, and only involves copying. This implies that some of the attention devoted to the input is harmful, and leads to inaccuracy, and slower training. This implies that for a given graph, there is at least a sub-quadratic attention that outperforms full attention. For example, if it would only consider the tokens that belong to the shortest path. The question is how can a more effective attention be developed? An interesting direction could come from inhibitory circuits. In the cortico-thalamus circuit that process visual information from the retina, there are two types of glutamatergic input (synapses): drivers, and modulators. While most of the information is delivered by driver synapses, there are ~10x more modulator synapses than drivers. More importantly, the information does not flow unidirectionally from the thalamus to the cortex, but rather bidirectionally through efferent copies. Feedback inhibitory cycles drive efficiency in the brain, by only focusing on the most salient visual stimulus ([S. Sherman, 2016 - U. Chicago](#)). Given that transformers function is primarily excitatory driven forward information, inhibitory circuits could lead to only delivering, and processing information that has been deemed relevant to the present prediction. In so doing, reducing entropy, and acquiring certainty to resemble predictions on accurate plans. Theoretically, training the inhibitory signal could also accelerate learning. Given that backpropagation currently only acts on driver circuits, it could instead train inhibitory neurons to select *when* information is delivered, without affecting *what* is delivered.